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Chapter 2

Predicting the Future: Variance and Sample Outcomes

Probability distributions that have values associated with the elements have two characteristics which, taken together, describe most of the behavior of the distribution for repeated trials. The first, described in the previous chapter, is expected value. The second is ***variance,*** a measure of the dispersion of the outcomes from the expectation. To characterize these two terms loosely, expected value measures how much you will win on average; variance measures how far your specific results may be from the expected value.

When we consider variance, we are attempting to capture the range of outcomes that can be expected from a number of trials. In many fields, the range of outcomes is of particular concern on both sides of the mean. For example, in many manufacturing environments there is a band of acceptability and outcomes on either side of this band are undesirable. In poker, there is a tendency to characterize variance as a one-sided phenomenon, because most players are unconcerned with outcomes that result in winning much more than expectation. In fact, "variance" is often used as shorthand for negative swings.

This view is somewhat practical, especially for professional players, but creates a tendency to ignore positive results and to therefore assume that these positive outcomes are more representative of the underlying distribution than they really are. One of the important goals of statistics is to find the probability of a certain measured outcome given a set of initial conditions, and also the inverse of this - inferring the initial conditions from the measured outcome. In poker, both of these are of considerable use. We refer to the underlying distribution of outcomes from a set of initial conditions as the ***population*** and the observed outcomes as the ***sample.*** In poker, we often cannot measure all the elements of the population, but must content ourselves with observing samples.

Most statistics courses and texts provide material on probability as well as a whole slew of sampling methodologies, hypothesis tests, correlation coefficients, and so on. In analyzing poker we make heavy use of probability concepts and occasional use of other statistical methods. What follows is a quick-and-dirty overview of some statistical concepts that are useful in analyzing poker, especially in analyzing observed results. Much information deemed to be irrelevant is omitted from the following and we encourage you to consult statistics textbooks for more information on these topics.

A commonly asked question *in* poker is "How often should I expect to have a winning session?" Rephrased, this question is "what is the chance that a particular sample taken from a population that consists of my sessions in a game will have an outcome > 0?" The most straightforward method of answering this question would be to examine the probability distribution of your sessions in that game and sum the probabilities of all those outcomes that are greater than zero.

Unfortunately, we do not have access to that distribution - no matter how much data you have collected about your play in that game from the past, all you have is a sample. However, suppose that we know somehow your per-hand expectation and variance in the game, and we know how long the session you are concerned with is. Then we can use statistical methods to estimate the probability that you will have a winning session. The first of these items, expected value (which we can also call the ***mean*** of the distribution) is familiar by now; we discussed it in Chapter 1.

Variance

The second of these measures, ***variance,*** is a measure of the deviation of outcomes from the expectation of a distribution. Consider two bets, one where you are paid even money on a coin

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flip, and one where you are paid 5 to 1 on a die roll, winning only when the die comes up 6. Both of these distributions have an EV of 0, but the die roll has significantly higher variance. $^1/_{16}$ of the time, you get a payout that is 5 units away from the expectation, while $^5/_6$ of the time you get a payout that is only 1 unit away from the expectation. To calculate variance, we first square the distances from the expected value, multiply them by the probability they occur, and sum the values.

For a probability distribution *P*, where each of the n outcomes has a value x_i and a probability p_i, then the variance of *P*, V_p is:

$$V_p = \sum_{i=1}^{n} p_i (x_i - < P >)^2 \qquad \textbf{(2.1)}$$

Notice that because each term is squared and therefore positive, variance is always positive. Reconsidering our examples, the variance of the coinflip is:

$V_C = (^1/_2)(1-0)^2 + (^1/_2)(-1-0)^2$
$V_C = 1$

While the variance of the die roll is:

$V_D = (^5/_6)(-1-0)^2 + (^1/_6)(5-0)^2$
$V_D = 5$

In poker, a loose-wild game will have much higher variance than a tight-passive game, because the outcomes will be further from the mean (pots you win will be larger, but the money lost in pots you lose will be greater). Style of play will also affect your variance; thin value bets and semi-bluff raises are examples of higher-variance plays that might increase variance, expectation, or both. On the other hand, loose-maniacal players may make plays that increase their variance while decreasing expectation. And playing too tightly may reduce both quantities. In Part IV, we will examine bankroll considerations and risk considerations and consider a framework by which variance can affect our utility value of money. Except for that part of the book, we will ignore variance as a decision-making criterion for poker decisions. In this way variance is for us only a descriptive statistic, not a prescriptive one (as expected value is).

Variance gives us information about the expected distance from the mean of a distribution. The most important property of variance is that it is directly additive across trials, just as expectation is. So if you take the preceding dice bet twice, the variance of the two bets combined is twice as large, or 10.

Expected value is measured in units of expectation per event; by contrast, variance is measured in units of expectation squared per event squared. Because of this, it is not easy to compare variance to expected value directly. If we are to compare these two quantities, we must take the square root of the variance, which is called the ***standard deviation.*** For our dice example, the standard deviation of one roll is $\sqrt{5} \approx 2.23$. We often use the Greek letter a (sigma) to represent standard deviation, and by extension σ^2 is often used to represent variance in addition to the previously utilized *V*.

$$\sigma = \sqrt{V} \qquad \textbf{(2.2)}$$
$$\sigma^2 = V \qquad \textbf{(2.3)}$$

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The Normal Distribution

When we take a single random result from a distribution, it has some value that is one of the possible outcomes of the underlying distribution. We call this a ***random variable.*** Suppose we flip a coin. The flip itself is a random variable. Suppose that we label the two outcomes 1 (heads) and 0 (tails). The result of the flip will then either be 1 (half the time) or 0 (half the time). If we take multiple coin flips and sum them up, we get a value that is the summation of the outcomes of the random variable (for example, heads), which we call a ***sample.*** The sample value, then, will be the number of heads we flip in whatever the size of the sample.

For example, suppose we recorded the results of 100 coinflips as a single number - the total number of heads. The expected value of the sample will be 50, as each flip has an expected value of 0.5.

The variance and standard deviation of a single flip are:

$$\sigma^2 = (\,^1/_2\,)(1 - \,^1/_2\,)^2 + (^1/_2\,)(0 - \,^1/_2\,)^2$$
$$\sigma^2 = \,^1/_4$$
$$\sigma = \,^1/_2$$

From the previous section, we know also that the variance of the flips is additiveю

So the variance of 100 flips is 25.

Just as an individual flip has a standard deviation, a sample has a standard deviation as well. However, unlike variance, standard deviations are not additive. But there is a relationship between the two.

For N trials, the variance will be:

$$\sigma^2{}_N = N\sigma^2$$
$$\sigma_N = \sigma\sqrt{N} \qquad \textbf{(2.4)}$$

The square root relationship of trials to standard deviation is an important result, because it shows us how standard deviation scales over multiple trials. If we flip a com once, it has a standard deviation of $^1/_2$. If we flip it 100 times, the standard deviation of a sample containing 100 trials is not 50, but 5, the square root of 100 times the standard deviation of one flip. We can see, of course, that since the variance of 100 flips is 25, the standard deviation of 100 flips is simply the square root, 5.

The distribution of outcomes of a sample is itself a probability distribution, and is called the ***sampling distribution.*** An important result from statistics, the ***Central Limit Theorem,*** describes the relationship between the sampling distribution and the underlying distribution. What the Central Limit Theorem says is that as the size of the sample increases, the distribution of the aggregated values of the samples converges on a special distribution called the ***normal distribution.***

The normal distribution is a bell-shaped curve where the peak, of the curve is at the population mean, and the tails asymptotically approach zero as the x-values go to negative or positive infinity. The curve is also scaled by the standard deviation of the population. The total area under the curve of the normal distribution (as with all probability distributions) is equal to 1, and the area under the curve on the interval $[x_1, x_2]$ is equal to the probability that a particular result will

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fall between x_1 and x_2. This area is marked region A in Figure 2.1.

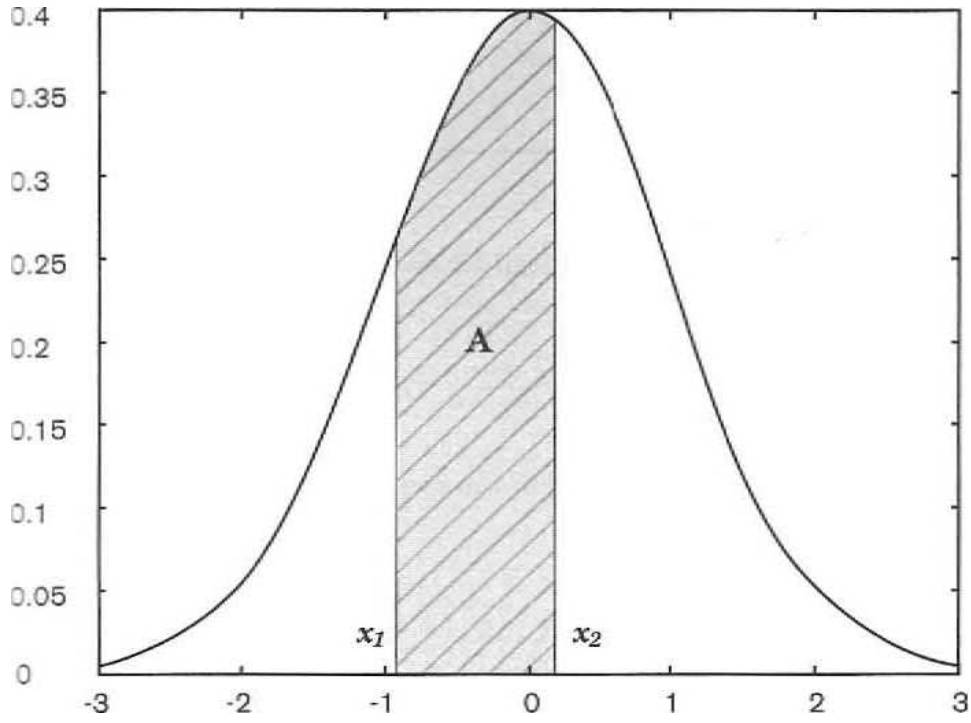

Figure 2.1. Std. Normal Dist, A = p(event between x, and

A little less formally, the Central Limit Theorem says that if you have some population and take a lot of big enough samples (how big depends on the type of data you're looking at), the outcomes of the samples will follow a bell-shaped curve around the mean of the population with a variance that's related to the variance of the underlying population.

The equation of the normal distribution function of a distribution with mean μ and standard deviation σ is:

$$N(x,\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{(x-\mu)^2}{2\sigma^2}\right) \qquad \textbf{(2.5)}$$

Finding the area between two points under the normal distribution curve gives us the probability that the value of a sample with the corresponding mean and variance will fall between those two points. The normal distribution is symmetric about the mean, so $^1/_2$ of the total area is to the right of the mean, and $^1/_2$ is to the left. A usual method of calculating areas under the normal curve involves creating for each point something called a ***z-score,*** where $z = (x-\mu)/\sigma$. This z-score represents the number of standard deviations that the outcome ***x*** is away from the mean.

$$z = (x-\mu)/\sigma \qquad \textbf{(2.6)}$$

We can then find something called the ***cumulative normal distribution*** for a z-score z**,** which is the area to the left of z under the curve (where the mean is zero and the standard deviation is 1). We call this function Φ(z). See Figure 2.2

If z is a normalized z-score value, then the cumulative normal distribution function for ***z*** is:

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$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{\to\infty}^{z} exp\left(-\frac{x^2}{2}\right) \qquad \textbf{(2.7)}$$

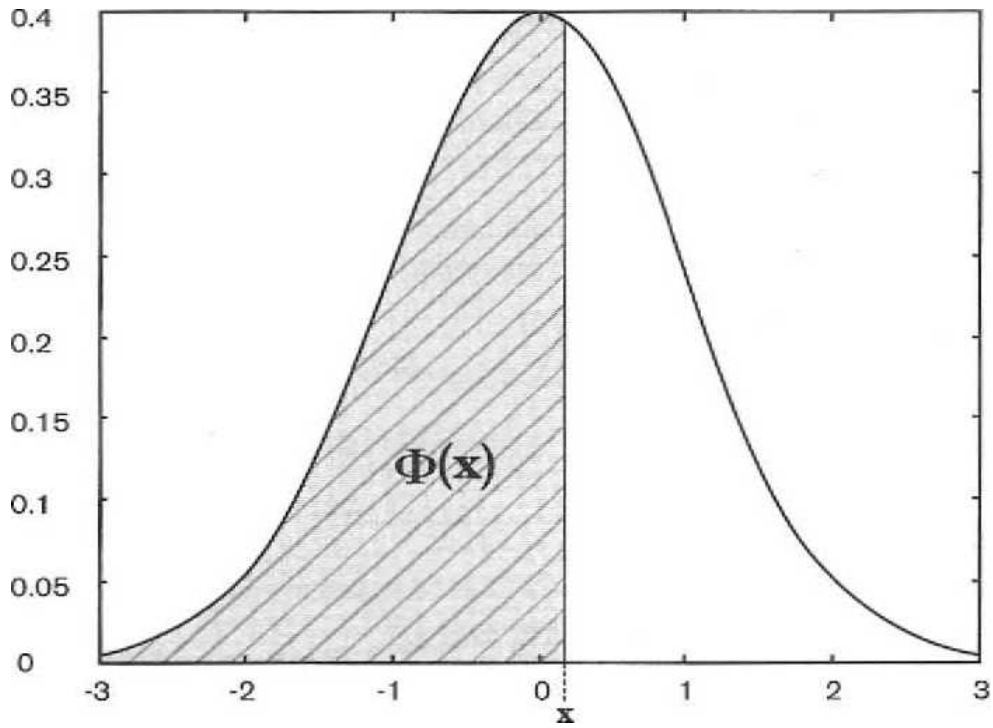

Figure 2.2, Cumulative normal distribution

Finding the area between two values x_1 and x_2 is done by calculating the z-scores z_1 and z_2 for x_1 and x_2, finding the cumulative normal distribution values $\Phi(z_1)$ and $\Phi(z_2)$ and subtracting them.

If $\Phi(z)$ is the cumulative normal distribution function for a z-score of z, then the probability that a sample taken from a normal distribution function with mean μ and standard deviation σ will fall between two z-scores x_1 and x_2 is:

$$p = \Phi\left(\frac{x_1-\mu}{\sigma}\right) - \Phi\left(\frac{x_2-\mu}{\sigma}\right) \qquad \textbf{(2.8)}$$

Statisticians have created tables of $\Phi(z)$ values and spreadsheet programs can derive these values as well. Some key values are well known to students of statistics. The area between z = -1 and z = +1 is approximately 0.68; the area between z = -2 and z = +2 is approximately 0.955, and the area between z = -3 and z= +3 is approximately 0.997.

These values mean that the value of a sample from a normal distribution will fall:

Between $(\mu - \sigma)$ and $(\mu + \sigma)$ of the mean 68% of the time.
Between $(\mu - 2\sigma)$ and $(\mu + 2\sigma)$ of the mean 95.5% of the time.
Between $(\mu - 3\sigma)$ and $(\mu + 3\sigma)$ of the mean 99.7% of the time.

An example may help to make this clear. Let us modify the parameters of the die roll game we discussed before. A die is thrown, and the player loses 1 unit on 1-5, but wins 6 units on a roll of

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6. We'll call this new game D_2.

The expectation of the game is:

$< D_2 > = ({}^5/_6)(-1) + ({}^1/_6)(6)$
$< D_2 > = {}^1/_6$ units /trial

When the player wins, he gains six units. Subtracting the mean value of ${}^1/_6$ from this outcome, we obtain:

$Vwin = (6 - {}^1/_6)^2$
$Vwin = ({}^{35}/_6)^2$

Likewise, when he loses he loses one unit. Subtracting the mean value of ${}^1/_6$ from this, we have:

$Vlose = (-1 - {}^1/_6)^2$
$Vlose = (- {}^7/_6)^2$

The variance of the game is:

$V_{D2} = p(\text{win})(V_{\text{win}}) + p(\text{lose})(V_{\text{lose}})$
$V_{D2} = ({}^1/_6)({}^{35}/_6)^2 + ({}^5/_6)(- {}^7/_6)^2$
$V_{D2} \approx 6.806$ units^2/trial^2

Suppose we toss the die 200 times. What is the chance that the player will win overall in 200 tosses? Win 40 units or more? Win 100 units or more?

We can solve this problem using the techniques we have just summarized. We first calculate the standard deviation, sometimes called the standard error, of a sample of 200 trials. This will be:

$\sigma = \sqrt{V} = \sqrt{6.806}$=2.61 units/trial

Applying equation 2.4 we get:

$\sigma_N = \sigma\sqrt{N}$
$\sigma_{200} = 2.61\sqrt{200} = 36.89$ units/200 trials

For 200 trials, the expected value, or mean (μ) of this distribution is ${}^1/_6$ units/trial times 200 trials or 33.33 units. Using Equation 2.6, we find the z-score of the point 0 as:

$z_x = (x-\mu)/\sigma$ where x=0
$z_0 = (0-33.33)/36.89$
$z_0 = -33.33/36.89$
$z_0 = -0.9035$

Consulting a probability table in a statistics textbook, we find that the probability that an observed outcome will lie to the left of this z-score, Φ(-0.9035) is 0.1831, Hence, there is an 18.31% chance that the player will be behind after 200 trials.

To find the chance of being 40 units ahead, we find that point's z-score:

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z_{40} = (40-33.33)/(36.89) = 0.1807
Φ(0,1807) = 0,5717

But Φ(0.1807) is the probability that the observed outcome lies to the left of 40 units, or that we lose at least 40 units. To find the probability that we are to the right of this value, or are ahead 40 units, we must actually find 1 - Φ(0.1807).

1 - Φ(0.1807) = 1 - 0.5717
= 0.4283

So there is a 42.83% chance of being ahead at least 40 units after 200 tosses.

And similarly for 100 units ahead:

z_{100} = (100-33.33)/(36.89) = 1.8070

From a probability table we find that Φ(1.8070) = 0.9646. Thus, for:

p = 1 - Φ(1.8070)
p = 0.0354

The probability of being 100 units ahead after 200 tosses is 3.54%.

These values, however, are only approximations; the distribution of 200-roll samples is not quite normal. We can actually calculate these values directly with the aid of a computer. Doing this yields:

	Direct Calculation	**Normal Approx**
Chance of being ahead after 200 trials:	81.96%	81.69%
Chance of being ahead at least 40 units:	40.46%	42.83%
Chance of being ahead at least 100 units:	4.44%	3,54%

As you can see, these values have slight differences. Much of this is caused by the fact that the direct calculations have only a discrete amount of values. 200 trials of this game can result in outcomes of +38 and +45, but not +39 or +42, because there is only one outcome possible from winning, a +6. The normal approximation assumes that all values are possible.

Using this method, we return to the question posed at the beginning of the chapter: "How often will I have a winning session?" To solve this problem, we need to know the player's expectation per hand, his variance per hand and the size of his sessions. Consider a player whose win rate in some particular game is approximately 0.015 bets per hand, and whose standard deviation over the same interval is 2 bets per hand. Assume this player plans to play sessions of 300 hands (these would be roughly nine-hour sessions in live poker; perhaps only two to three hours online). How often should this player expect to win (have a result greater than 0) in a session?

First we can calculate the expected value of a sample, μ_N:

μ_N = Nμ
μ_N = (300)(0.015)
μ_N = 4.5

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Second, we calculate his standard deviation for a 300-hand session.

$\sigma_N = \sigma\sqrt{N}$
$\sigma_{300} = (2)(\sqrt{300})$
$\sigma_{300} = 34.6$

Next, we find the z-score and Φ(z) for this value:

$z_x = (x-\mu_N)/\sigma$
$z_0 = (0 - 4.5)/34.6$
$z_0 = -0.1299$

From a probability table we find:

$\Phi(-0.1299) = 44.83\%$
$p = 1 - \Phi(-0.1299)$
$p = l - 0.4483$
$p = 0.55171$

This indicates that the player has a result of 0 or less 44.83% of the time - or correspondingly has a winning session 55.17% of the time. In reality, players may change their playing habits in order to produce more winning sessions for psychological reasons, and in reality "win rate" fluctuates significantly based on game conditions and for other reasons. But a player who simply plays 300 hands at a time with the above performance metrics should expect to be above zero just over 55% of the time.

A more extreme example: A player once told one of the authors that he had kept a spreadsheet where he marked down every AQ vs. AK all-in preflop confrontation that he saw online over a period of months and that after 2,000 instances, they were running about 50-50.

How likely is this, assuming the site's cards were fair and we have no additional information based on the hands that were folded?

Fust, let's consider the variance or standard deviation of a single confrontation.

AK vs. AQ is about 73.5% all-in preflop (including all suitedness combinations). Let's assign a result of 1 to AK winning and 0 to AQ winning. Then our mean is 0.735. Calculating the variance of a single confrontation:

$V = (0.735)(1- 0.735)^2 + (0.265)(0- 0.735)^2$
$V = 0.1948$
$\sigma = 0.4413$

The mean of 2,000 hands is:

$\mu_N = N\mu$
$\mu_N = (2000)(0.735)$
$\mu_N = 1470$

For a sample of 2,000 hands, using Equation 2.4, the standard deviation is:

$\sigma_N = \sigma\sqrt{N}$

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$\sigma_{2000} = (0.4413)(\sqrt{2000})$
$\sigma_{2000} = 19.737$

The result reported here was approximately 50% of 2000, or 1000, while the sample mean would be about 1470 out of 2000. We can find the z-score using Equation 2.6:

$z_x = (x-\mu_N)/\sigma$
$z_{1000} = (1000-1470)/(19.737)$
$z_{1000} = -23.815$

The result reported in this case was 1000 out of 2000, while the expected population mean would be 1470 out of 2000. This result is then about 23.815 standard deviations away from the mean. Values this small are not very easy to evaluate because they are so small - in fact, my spreadsheet program calculates $\Phi(-23.815)$ to equal exactly zero. Suffice it to say that this is a staggeringly low probability.

What's likely is that in fact, this player was cither exaggerating, outright lying, or perhaps made the common mistake of forgetting to notice when AK beat AQ, because that is the "expected" result. This is related to a psychological effect known as "perception bias" - we tend to notice things that are out of the ordinary while failing to notice things that are expected. Or perhaps the online site was in fact "rigged." When this example was posed to a mailing list, the reaction of some participants was to point to the high variance of poker and assert that a wide range of outcomes is possible in 2000 hands. However, this confuses the altogether different issue of win/loss outcomes in 2000 poker hands (which docs have quite high variance) with the outcome of a single poker hand (which has far less). The variance of a single hand in terms of big bets in most forms of poker is much higher than the variance of the winner of an all-in preflop confrontation. One lesson of this example is not to confuse the high variance of poker hand dollar outcomes with the comparatively low variance of other types of distributions.

When we play poker, many random events happen. We and the other players at the table are dealt random cards taken from a distribution that includes all two-card combinations. There is some betting, and often some opportunity to either change our cards, or to have the value of our hand change by the dealing of additional cards, and so on. Each hand results in some outcome for us. whether it is whining a big pot, stealing the blinds, losing a bet or two, or losing a large pot. This outcome is subject to all the smaller random events that occur within the hand, as well as events that are not so random - perhaps we get a tell on an opponent that enables us to win a pot we would not otherwise have, or save a bet when we are beaten. Nevertheless, the power of the Central Limit Theorem is that outcomes of individual hands function approximately as though they were random variables selected from our "hand outcomes" distribution. And likewise, outcomes of sessions, and weeks, and months, and our whole poker career, behave as though they were appropriately-sized samples taken from this distribution.

The square root relationship of trials to standard deviation makes this particularly useful, because as the number of trials increases, the probability that our results will be far away from our expected value in relative terms decreases.

Assume we have a player whose distribution of hand outcomes at a particular limit has a mean of S75 per 100 hands, with a variance of $6,400 per hand. If we sample different numbers of hands from this player's distribution, we can see how the size of the sample impacts the dispersion of the results. We know that the probability that this player's results for a given sample will be between the mean minus two standard deviations and the mean plus two standard deviations is 95.5%. We will identify for each sample size:

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- The mean μ_N
- The standard deviation σ
- The two endpoints of the 95.5% probability interval.

Hands	μ_N	σ	Lower endpoint	Higher endpoint
100	$75	$800.00	($1,525.00)	$1,675.00
500	$375.00	$1,788.85	($3,202.71)	$3,952.71
1,000	$750.00	$2,529.82	($4,309.64)	$5,809.64
5,000	$3,750.00	$5,656.85	($7,563.71)	$15,063.71
25,000	$18,750.00	$12,649.11	($6,548.22)	$44,048.22
50,000	$37500.00	$17888.54	$1,722.91	$73,277.09
100,000	$75,000.00	$25,298.22	$24,403.56	$125,596.44
1,000,000	$750,000.00	$80,000.00	$590.000.00	$910,000.00

As you can see, for smaller numbers of hands, outcomes vary widely between losses and wins. However, as the size of the sample grows, the relative closeness of the sample result becomes larger and larger - although its absolute magnitude continues to grow. Comparing the standard deviation of one million hands to the standard deviation for one hundred hands, the size of a standard deviation is a hundred times as large in absolute terms, but more than a hundred times smaller relative to the number of hands. This is the ***law of large numbers*** at work; the larger the sample, the closer on a relative basis the outcomes of the sample will be.

Key Concepts

- Variance, the weighted sum of the squares of the distance from the mean of the outcomes of a distribution, is a valuable measurement of dispersion from the mean.
- To compare variance to expected value, we often use its square root, standard deviation.
- Variance is additive across trials. This gives rise to the square root relationship between the standard deviation of a sample of multiple trials and the standard deviation of one trial.
- The Central Limit Theorem tells us that as the size of a sample increases, the distribution of the samples behaves more and more like a normal distribution. This allows us to approximate complex distributions by using the normal distribution and to understand the behavior of multiple-trial samples taken from a known distribution.

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